

- 7 The curve C has equation $y = f(x)$, where $f(x) = \frac{x^2}{x+1}$.

(a) Find the equations of the asymptotes of C . [3]

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(b) Find the coordinates of any stationary points on C . [2]

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(c) Sketch C .

[3]

(d) Find the coordinates of any stationary points on the curve with equation $y = \frac{1}{f(x)}$. [2]

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- (e) Sketch the curve with equation $y = \frac{1}{f(x)}$ and find, in exact form, the set of values for which

$$\frac{1}{f(x)} > f(x).$$
[6]

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